coursera | Northeastern University

Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Getting Started

- Course Introduction
  Video • 1 min
- Course Overview
  Reading • 2 min
- Syllabus
  Reading • 10 min
- Meet Your Instructor
  Video • 1 min
- Meet Your Fellow Learners
  Discussion Prompt • 10 min
- Academic Integrity
  Reading • 1 min

Lesson 1: Introduction to Linear Algebra

- **Introduction to Machine Learning**
  Reading • 25 min
- Introduction to Linear Algebra
  Reading • 4 min
- Why Linear Algebra and Mathematics?
  Reading • 2 min
- Notation
  Reading • 5 min

Lesson 2: Systems of Linear Equations

- Foundational Concepts of Systems of Linear Equations
  Reading • 20 min
- [H5P] System of Linear Equations
  Ungraded App Item • 20 min
- Solved Example and Recommended Resources
  Reading • 20 min
- Check Your Knowledge: System of Linear Equations
  Graded Assignment • 16 min

Lesson 3: Matrices

- Matrices and Matrix Operations
  Reading • 20 min
- Matrices
  Video • 8 min
- Helpful Resources and Solved Examples
  Reading • 10 min
- Check Your Knowledge: Matrices
  Graded Assignment • 16 min

Lesson 4: Vector Spaces

- Foundations of Vector Spaces: Operations and Subspaces
  Reading • 20 min
- Vector Space
  Video • 7 min
- Vector Space Properties
  Reading • 10 min
- Subspace
  Reading • 1 min
- Check Your Knowledge: Vector Spaces
  Graded Assignment • 14 min

Lesson 5: Orthogonal Complement and Projection

- Introduction to Orthogonal Complement
  Reading • 20 min
- Introduction to Orthogonal Projections
  Reading • 40 min

Introduction to Machine Learning

0:03 / 0:46 2x

Read Chapter 1: “Introduction and Motivation” from Mathematics for Machine Learning by Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong.

In this chapter, you will be introduced to the fundamental concepts of machine learning, including the importance of data, the role of models, and the process of learning through optimization. You will also explore the significance of creating general-purpose methodologies and the importance of understanding the mathematical foundations of machine learning.

As you read, consider the following questions:

1. How can understanding the mathematical foundations of machine learning help in creating and debugging new models?
2. Why is it important for machine learning models to generalize well to new, unseen data?

Go to next item ✓ Completed

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